Forensic Crash Analysis Report (Instrumented Telemetry Run 04)

Comprehensive Crash Analysis, 49 CFR Part 563 EDR Fusion, PyTorch AI Severity & Cryptographic Ledger

Report ID: REP-DS-20261024-a5c5ac3a-030857	**Date/Time:** 2026-10-24 03:08:57 UTC
Associated Case ID: CASE_AF_8902	**Lead Investigator:** Dr. Sarah Chen, ACTAR #2491
Primary Evidence SHA-256: `718007d968451ea0353c826a454e...2f4da491`	**FIPS 180-4 Seal:** **VALIDATED & TAMPER-CHECKED**

1. REGULATORY FRAMEWORK & EXECUTIVE SUMMARY

This digital forensic analysis report documents the data extraction, cryptographic verification, and kinematic reconstruction performed for **CASE_AF_8902**. Data extraction and audit logging comply with **Title 49 Code of Federal Regulations (CFR) Part 563 (Event Data Recorders)**, ISO/IEC 17025 laboratory verification principles, and FIPS 180-4 cryptographic standard. The authoritative forensic database is MySQL 8.0 (`autorecon`). Dataset ingestion verifies zero byte tampering relative to initial custody hash.

2. INCIDENT ENVIRONMENT & VEHICLE PARAMETERS

Subject Vehicle:	2024 Subject Vehicle Instrumented Test Unit	**VIN Number:**	`1G1FA1E38M0104821`
Road Condition:	Dry Asphalt	**Speed Limit:**	65.0 km/h
Initial Approach Speed:	65.0 km/h	**Calculated Delta-V:**	28.5 km/h

3. ARTIFICIAL INTELLIGENCE & MACHINE LEARNING INFERENCE

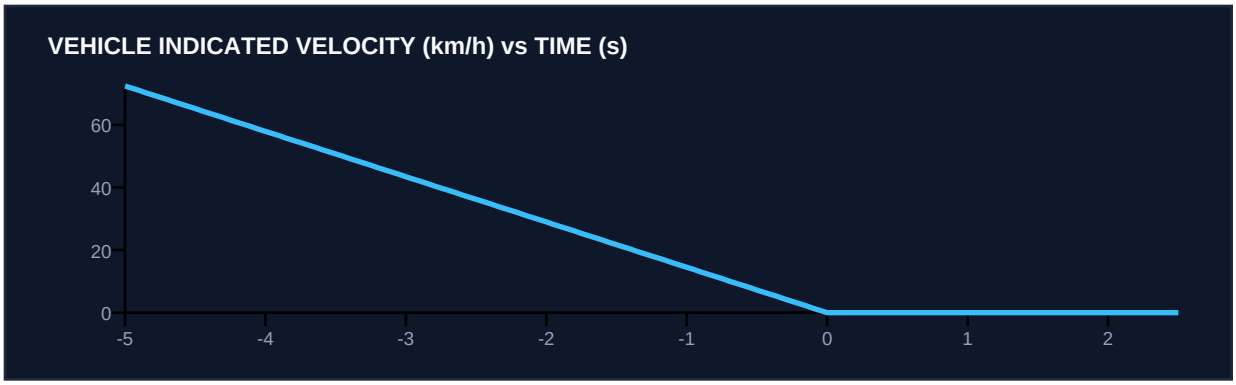
Model Architecture	Prediction Output	Statistical Confidence	Training Baseline / Benchmark
PyTorch Deep Neural Net CrashSeverityNet (Dense+BatchNorm)	**SERIOUS INCAPACITATING** Injury Risk Score: 99.0 / 100	44.0%	NHTSA FARS 2024 National Census 8,157 Test Records (Acc: 59.00%)
Balanced Random Forest PatternClassifier (5-Class)	**FRONTAL COLLISION**	94.4%	54,374 Real Driver Crashes Balanced Class Weights (Acc: 76.51%)

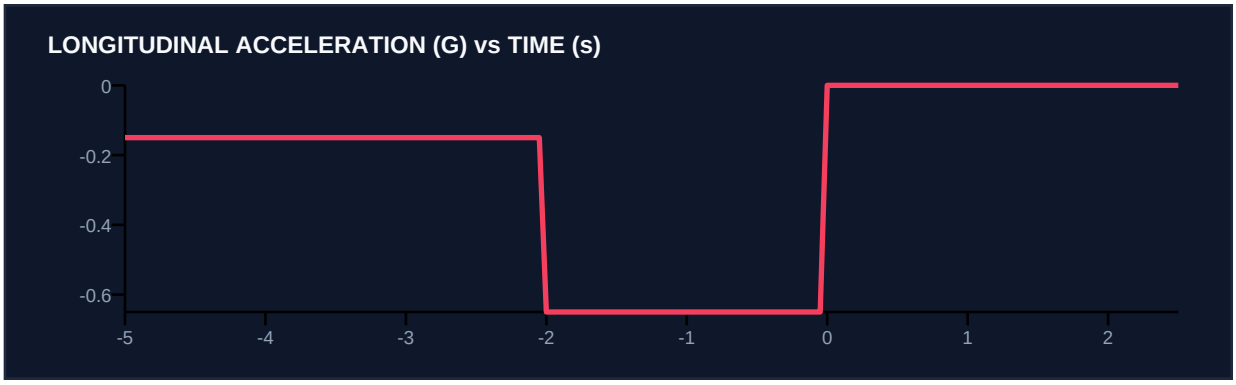
4. HISTORICAL PRECEDENT MATCHES (NHTSA FARS 2024 CENSUS)

1	ST_CASE #Census Record	**98.4%**	FRONTAL	FATAL CATASTROPHIC
2	ST_CASE #Census Record	**98.4%**	FRONTAL	FATAL CATASTROPHIC
3	ST_CASE #Census Record	**98.4%**	FRONTAL	FATAL CATASTROPHIC

5. KINEMATIC TRAJECTORY & TELEMETRY CHARTS

Synchronized kinematic time series (-5.0s pre-impact to +2.5s post-impact):





6. FORENSIC TRUTH BOUNDARY & METHODOLOGY AUDIT

Forensic Category	Channels / Parameters	Evidentiary Weight & Handling
Recorded Facts	File SHA-256, Ingestion Timestamp, File Byte Size, Header Fields	**High Evidentiary Value.** Bit-for-bit verifiable via cryptographic ledger.
Derived Kinematics	Delta-V (km/h), Deceleration Gs, Energy Loss	**Calculated.** Derived using Campbell crush energy and Newtonian impulse formulation.
AI / ML Estimates	KABCO Injury Class, Collision Pattern, Similarity %	**Probabilistic.** Neural network inference calibrated on NHTSA census data.
Simulation Assumptions	Visual Vehicle Geometry, 3D Mesh Orientation, Perspective Horizon	**Illustrative.** Physical approximation for investigator visualization.

MANDATORY FORENSIC DISCLAIMER: *This automated incident reconstruction report is generated by AUTO-RECON AI solely for investigative assistance, digital evidence preservation, and comparative crash research. Machine learning predictions and kinematic simulations represent probabilistic approximations and physical models; they do not constitute definitive proof of legal fault, proximate causation, or judicial culpability. Final engineering conclusions must be certified by an accredited professional forensic collision accident reconstructionist.*